

WEEK 3

Statistical Modeling & Hypothesis Testing

Statistical Modeling of Retail Order Value

Investigating the influence of order characteristics, product diversity, geographic and time-related factors on retail order value.

Internship	Junior Data Scientist Internship — Yuva Intern
Intern	Anand Raj Yadav
Week	3 of 6
Dataset	Online Retail — cleaned Week 1 dataset
Model	OLS regression with HC3 robust inference
Observation Unit	Invoice / order level

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Data Science • Statistics • Business Analytics

1. Executive Summary

This study develops and evaluates a statistical regression model using the cleaned Online Retail dataset from the previous internship weeks. The workflow moves from invoice-level feature engineering and hypothesis formulation to regression estimation, assumption diagnostics, robust inference, and sensitivity analysis.

The modeling dataset contains **19,960 invoice-level orders**. The dependent variable is **LogOrderValue**, a log-transformed order value used because raw order value was strongly right-skewed. The OLS model explains approximately **28.55%** of variation in the transformed target, with an adjusted R^2 of **28.49%**. The overall model is statistically significant ($p < 0.001$).

The Breusch–Pagan test indicated heteroscedasticity ($p < 0.001$), so HC3 heteroscedasticity-consistent standard errors were used for final coefficient inference. Among the three pre-specified hypotheses, **Unique Products** showed strong evidence of association with order value ($p = 4.235 \times 10^{-3}$), while Total Quantity ($p = 0.08383$) and Country ($p = 0.16141$) did not reach the 5% threshold.

Sensitivity analysis removed 701 potentially influential observations, reducing the sample from 19,960 to 19,259 orders. R^2 increased from **0.2855 to 0.4139**, demonstrating that influential orders materially affect model fit. These observations were not automatically treated as errors because unusually large retail orders can represent genuine commercial activity.

Overall conclusion: the study provides evidence that product diversity is associated with retail order value, while the tested quantity and country hypotheses are not supported at $\alpha = 0.05$. The results are associative, not causal, and provide a foundation for future predictive modeling and controlled business experimentation.

2. Research Objective and Hypotheses

Objective. To develop a statistical model that explains variation in retail order value and evaluates the significance, stability, and practical interpretation of the modeled relationships.

Research Question. Which order, product, geographic, and time-related characteristics are significantly associated with retail order value?

Hypothesis	Statement	Result
H1 — Total Quantity	H ₀ : no significant association. H _a : significant association.	Fail to Reject H ₀ p = 0.08383
H2 — Unique Products	H ₀ : no significant association. H _a : significant association.	Reject H ₀ p = 4.235 × 10 ⁻³
H3 — Country	H ₀ : country does not significantly contribute. H _a : country contributes significantly.	Fail to Reject H ₀ p = 0.16141

The 5% significance level ($\alpha = 0.05$) was used. Failing to reject H₀ does not prove that no relationship exists; it indicates insufficient evidence to reject the null under the specified model.

Statistical principle: statistical significance is not the same as practical significance, and association is not causation. The dataset is observational.

3. Data Preparation and Modeling Dataset

The Week 1 cleaned Online Retail dataset was used as the starting point. Positive sales transactions (Quantity > 0 and UnitPrice > 0) were retained for the primary order-value model. Negative-quantity records, which may represent returns, cancellations, or reversals, were excluded from the primary positive-sales target.

Line-level revenue was calculated as Quantity × UnitPrice. Transactions were then aggregated by InvoiceNo to create the invoice-level modeling dataset.

Feature	Construction / Role
OrderValue	Sum of line-level revenue within an invoice; target before transformation.
LogOrderValue	$\log(1 + \text{OrderValue})$; primary regression target.
TotalQuantity	Total quantity purchased within an invoice.
UniqueProducts	Number of distinct products within an invoice.
AvgUnitPrice	Mean UnitPrice within an invoice.
Country	Categorical geographic predictor; rare countries grouped as Other.
Hour / DayOfWeek / Month	Time-derived predictors from InvoiceDate.
IsWeekend	Binary weekend indicator.

Modeling dataset: 19,960 invoice-level orders. CustomerID was not used as a raw numerical predictor because it is an identifier, not a meaningful continuous measurement. Missing customer IDs were not artificially inferred.

4. Distribution, Assumptions and Diagnostics

Raw OrderValue was right-skewed: mean ≈ 533.17 , median ≈ 303.30 , and maximum = 168,469.60. This motivated the log transformation used for the dependent variable.

Diagnostic	Purpose	Result / Treatment
Distribution	Assess target suitability.	Strong right skew; LogOrderValue used.
VIF	Assess multicollinearity.	Used to screen relationships among predictors.
Q-Q plot	Assess residual normality.	Used as a graphical diagnostic.
Residual vs fitted	Assess systematic patterns.	Used to evaluate specification and variance behavior.
Breusch–Pagan	Test constant residual variance.	LM $p < 0.001$; F $p < 0.001 \rightarrow$ heteroscedasticity evidence.
HC3 robust SE	Improve inference under heteroscedasticity.	Used for final p-value interpretation.
Cook's Distance	Screen influential observations.	701 observations exceeded the screening threshold.

Why HC3: heteroscedasticity can make conventional OLS standard errors unreliable. HC3 provides heteroscedasticity-consistent standard errors while retaining the OLS coefficient estimates.

5. OLS Regression Model and Performance

The OLS specification modeled LogOrderValue using TotalQuantity, UniqueProducts, AvgUnitPrice, Hour, DayOfWeek, Month, IsWeekend, and CountryGroup. CountryGroup was represented through categorical indicators.

Model form: $\text{LogOrderValue} = \beta_0 + \beta_1(\text{TotalQuantity}) + \beta_2(\text{UniqueProducts}) + \beta_3(\text{AvgUnitPrice}) + \beta_4(\text{Hour}) + \beta_5(\text{DayOfWeek}) + \beta_6(\text{Month}) + \beta_7(\text{IsWeekend}) + \text{Country effects} + \varepsilon$

Metric	Value	Interpretation
Observations	19,960	Invoice-level observations in the full model.
R ²	0.285469	≈28.55% of variation in LogOrderValue explained.
Adjusted R ²	0.284860	Fit after accounting for model complexity.
MAE	0.712313	Error on log-transformed target scale.
RMSE	1.019302	Larger errors receive more weight; log scale.
Overall model p-value	< 0.001	Strong evidence that the model is jointly informative.

R² caution: 0.2855 is not “28.55% accuracy”; it means the predictors explain about 28.55% of variation in the transformed target.

Error-scale caution: MAE and RMSE are on the log scale and are not direct currency-unit errors.

6. Hypothesis Testing Results

Because heteroscedasticity was detected, final inference relies on HC3 robust standard errors.

Hypothesis	p-value	Decision	Interpretation
H1 — Total Quantity	0.08383	Fail to Reject H_0	Insufficient evidence of a conditional association at $\alpha = 0.05$.
H2 — Unique Products	4.235×10^{-3}	Reject H_0	Very strong evidence that product diversity is associated with order value.
H3 — Country	0.16141	Fail to Reject H_0	Insufficient evidence of a significant country contribution under this model.

Primary finding: UniqueProducts is the strongest of the three pre-specified hypothesis areas in statistical evidence.

Correct interpretation of non-significance: $p = 0.08383$ for Total Quantity and $p = 0.16141$ for Country do not prove that these variables have no effect. They indicate that this study does not have sufficient evidence to reject the respective null hypotheses at the 5% level.

Practical caution: even an extremely small p-value does not by itself establish a large business effect or causality.

7. Sensitivity Analysis and Robustness

Cook's Distance was used to screen potentially influential observations. The same regression specification was refitted after excluding observations above the screening threshold.

Measure	Full Model	Sensitivity Model
Observations	19,960	19,259
Removed	—	701
R ²	0.285469	0.413878

The increase in R² from 0.2855 to 0.4139 shows that the screened observations materially influence model fit. This is a substantive modeling insight: unusual retail orders can have disproportionate leverage on a regression model.

The observations were not automatically labeled as erroneous. In retail, unusually large orders may represent genuine bulk purchases, business customers, exceptional product combinations, or other real commercial activity.

Future modeling direction: robust regression, quantile regression, nonlinear methods, or customer-level hierarchical approaches may better capture heterogeneous retail behavior.

8. Business Implications and Recommendations

Finding	Potential Meaning	Recommended Follow-up
Unique product count is significant	Basket diversity is associated with order-value variation.	Test cross-selling, complementary products, bundles and recommendations.
Quantity not significant at 5%	Quantity alone lacks sufficient conditional evidence in this specification.	Test nonlinear quantity effects, product mix, discounts and interactions.
Country not significant at 5%	Geography alone does not add sufficient evidence after controls.	Explore finer customer/product segments rather than country-only strategies.
701 influential observations	A subset of orders materially affects model fit.	Review high-value orders as potential genuine business segments.

Concrete example: if product diversity remains predictive in future validation, a retailer could test product recommendations or complementary bundles designed to increase relevant products purchased together. A controlled experiment would be required to determine whether such an intervention causes higher order value.

Future data-science use: statistically informative features can become candidates for customer-value prediction, segmentation, recommendation, or demand forecasting. Future predictive models should use train/test validation and cross-validation.

9. Assumptions, Limitations and Future Improvements

Area	Consideration
Linearity	OLS assumes approximately linear conditional relationships; nonlinear effects may remain.
Independence	Orders are treated as independent, although repeated purchases by a customer may introduce dependence.
Heteroscedasticity	Breusch–Pagan evidence was significant; HC3 robust SE were therefore used.
Residual normality	Assessed graphically; deviations should be interpreted in the context of the large sample.
Multicollinearity	VIF was used to screen predictor relationships.
Observational design	Associations cannot establish causality; promotions, discounts and intent are not fully observed.
Customer information	CustomerID is missing for a portion of orders and was not used as a raw numeric predictor.
Returns / reversals	Negative-quantity records were excluded from the primary positive-sales model and should be analyzed separately.

Future improvements: add product categories and promotion/discount variables; engineer customer recency-frequency-monetary features; test nonlinear and interaction effects; compare Ridge/Lasso and tree-based models; use cross-validation; and investigate robust, quantile, or mixed-effects approaches.

10. Conclusion

This Week 3 study demonstrates a complete statistical modeling workflow on retail transaction data: invoice-level aggregation, feature engineering, explicit hypotheses, distribution assessment, multicollinearity screening, OLS estimation, residual diagnostics, robust inference, and sensitivity analysis.

The overall model was statistically significant and explained approximately 28.55% of variation in LogOrderValue. The strongest hypothesis-level evidence was associated with **UniqueProducts**. The available evidence did not support the Total Quantity or Country hypotheses at $\alpha = 0.05$.

Sensitivity analysis was especially informative: removing 701 potentially influential observations increased R^2 from 0.2855 to 0.4139. This demonstrates that unusual orders materially affect model fit and reinforces the importance of robust diagnostics and business-aware treatment of outliers.

The study therefore provides a statistically disciplined foundation for future predictive modeling and experimentation. Significant results identify associations worth investigating; non-significant results indicate insufficient evidence rather than proof of no effect; and model diagnostics are essential before drawing business conclusions.

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